

Rethinking long-context VLM as multi-view contrastive learning

Chau Truong

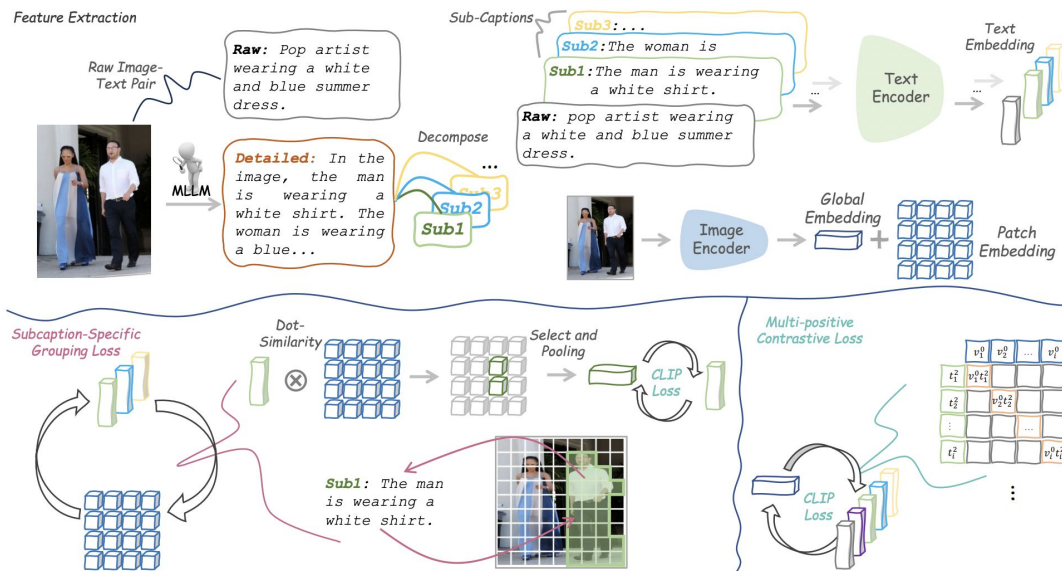
Sampling strategy – Stars and Bars



An outdoor daytime shot of a sculpted statue of George Washington in the foreground to the left and the University of Texas Clock Tower in the background to the right. A blue sky with two clouds floating above is seen behind the tower. The statue is highlighted in mint green verdigris on the highest points of the face, jacket trim, jacket buttons, and hood depicted on the statue. The majority of the statue is dark pewter. The statue is on a square block platform made of the same metal. The stone foundation has wording etched to it professionally. The font reads," GEORGE / WASHINGTON / WHEN / MADE / COMMANDER / IN / CHIEF / OF / THE / AMERICAN / ARMY / OF / THE / REVOLUTION / ON / JULY / 3 / 1775."

N stars - 4 Bars!

Multi-Positive loss from DreamLIP



$$\mathcal{L}_{\text{MPCL}}^{t2v} = - \sum_{i=1}^N \sum_{j=1}^K \log \frac{\exp(\cos\langle \mathbf{v}_i, \mathbf{t}_{i,j} \rangle / \tau)}{\sum_{n=1}^N \exp(\cos\langle \mathbf{v}_n, \mathbf{t}_{i,j} \rangle / \tau)}, \quad (5)$$

DreamLIP: Language-Image Pre-training with Long Captions (ECCV 2024)

Adaptive Multi-positive loss

$$\mathcal{L}_{t2v}^{\text{adaptive-MPCL}} = - \sum_{i=1}^N \sum_{j=1}^K W_{i,j} \log \left(\frac{\exp(\cos(v_i, t_{i,j}) / \tau)}{\sum_{n=1}^N \exp(\cos(v_n, t_{i,j}) / \tau)} \right).$$

$$W_{i,j} = \text{softmax}_j(\cos(v_i, t_{i,j})) = \frac{\exp(\cos(v_i, t_{i,j}))}{\sum_{k=1}^j \exp(\cos(v_i, t_{i,k}))}.$$

Sparsemax version are more suitable for noisy long-text image

$$W_{i,j} = \text{sparsemax}_j(\cos(v_i, t_{i,j})).$$

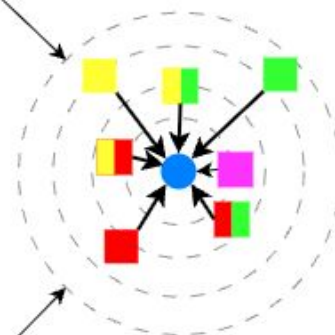
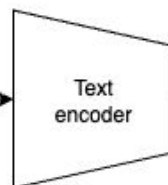
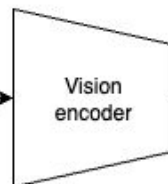
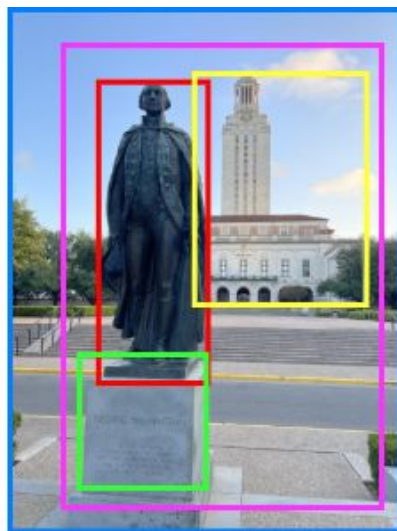
$$\mathcal{L}_{\text{learnable_MPCL}}^{v2t} = - \sum_{i=1}^N \sum_{j=1}^K \lambda_{i,j} \log \frac{\exp(\cos(\mathbf{v}_i, \mathbf{t}_{i,j}) / \tau)}{\sum_{n=1}^N \exp(\cos(\mathbf{v}_i, \mathbf{t}_{n,j}) / \tau)}.$$

→ $\mathbf{v}$, $\mathbf{t}$ receive no gradient
 $W_{\{i, j\}}$ updated by $\log(i, j)$

In-domain performance increase
 by 1% in the final epoch
 compared to learnable version

$$\mathcal{L} = 1/2 * (\mathcal{L}_{\text{learnable_MPCL}}^{v2t} + \mathcal{L}_{\text{learnable_MPCL}}^{i2t})$$

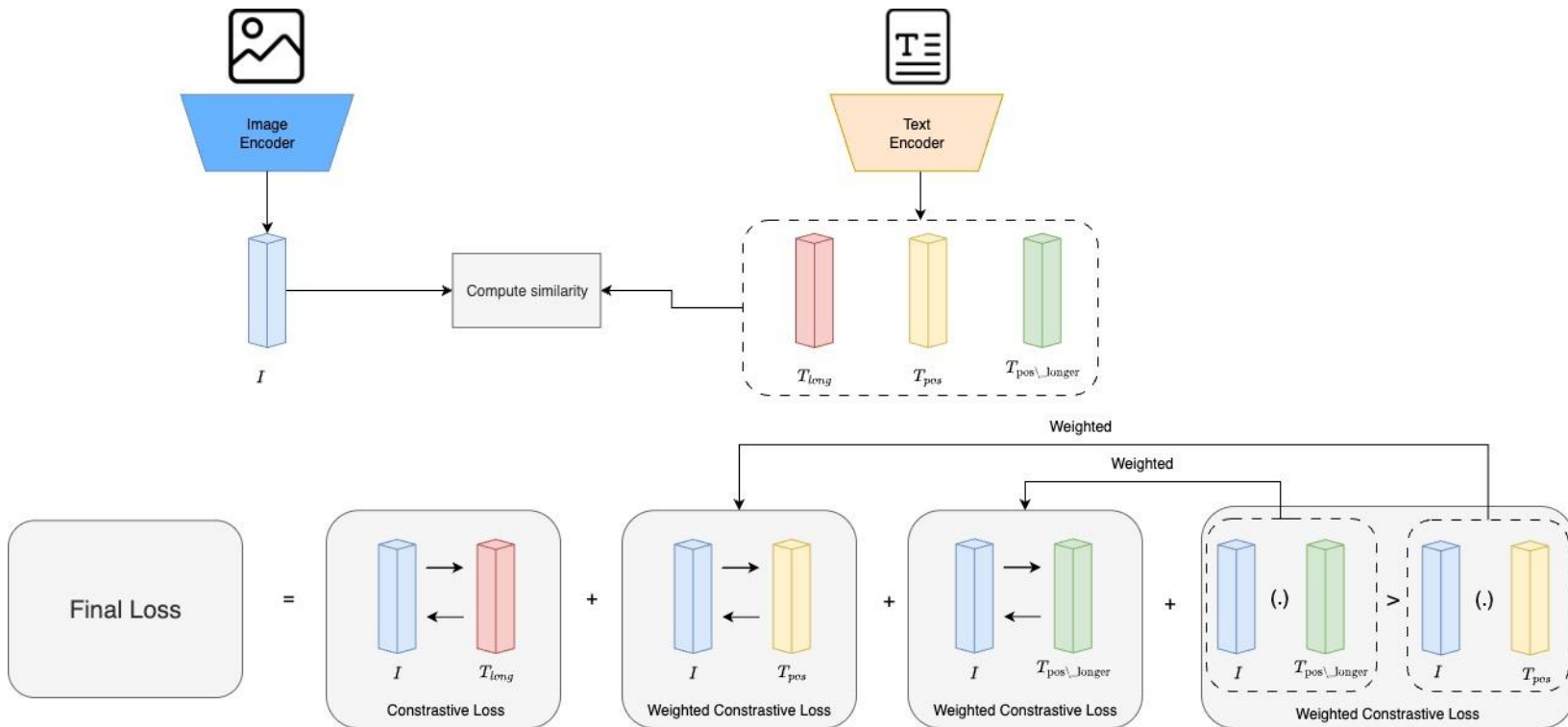
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Our methods

An outdoor daytime shot of a sculpted statue of George Washington in the foreground to the left and a building clock tower in the background to the right. The sky is bright blue with a few scattered clouds.. The tower resembles the University of Texas Tower, with a light beige facade, reddish-orange roof tiles, and a golden clock face.. The statue is highlighted in mint green verdigris on the highest points of the face, jacket trim, jacket buttons, and hood depicted on the statue. The majority of the statue is dark pewter. The statue is on a square block platform made of the same metal. The stone foundation has wording etched to it professionally. The font reads," GEORGE / WASHINGTON / WHEN / MADE / COMMANDER / IN / CHIEF / OF / THE / AMERICAN / ARMY / OF / THE / REVOLUTION / ON / JULY / 3 / 1775."

New version: Rethinking long-context VLM as multi-view contrastive learning



base correspondence - longer correspondence; optimized constraint